

# Lecture 3: Elasticities & Taxes — the Supply & Demand model II (Ch. 2 recap · §6.4)

*Applied Microeconomics — Master Notes · by Jakob V. Stangel*

**Required reading:** Friberg, §6.4 Elasticities (own-price, cross-price, income; supply elasticity) — building on the **Chapter 2** supply-and-demand model (see **Lecture 2**). *The lecture deck also teaches taxes on goods & services — that part draws on the textbook's §7.5 (unit tax, tax incidence), covered here because the deck foregrounds it.*

**Theme of the lecture** (Butera — from the deck, titled "*Supply and Demand model*"). Chapter 2 taught us to **move the curves around**; this lecture adds **precision**. The *"shape"* of a demand or supply curve — its *elasticity* — decides how much a shock (or a tax) shows up as a change in price versus a change in quantity. The deck's three goals: (1) understand what demand & supply elasticities mean, (2) derive them from demand/supply curves, and (3) use them to analyse the **effects of taxes**.

★ **The heart of the lecture:** elasticity = % responsiveness — "by what % does quantity change when price rises 1%." It's a **unit-free** number that summarises a curve's **shape**, so it's comparable across markets and time. The **shape decides how a shock splits into price vs quantity**: with **inelastic** demand a shock/tax hits **price**; with **elastic** demand it hits **quantity**. And a **tax** drives a **wedge** between the price buyers pay and sellers receive — **who bears it depends on the relative elasticities**, and (in perfect competition) **it doesn't matter who is legally taxed**.

**The red thread.** For any market ask: *how responsive is quantity to price (elastic or inelastic)?* — because that single question answers **how a supply shock moves the market**, **how a firm should price**, and **who really pays a tax**. Elasticity is the "shape" number that ties all three together.

## Key concepts / "modes" to use

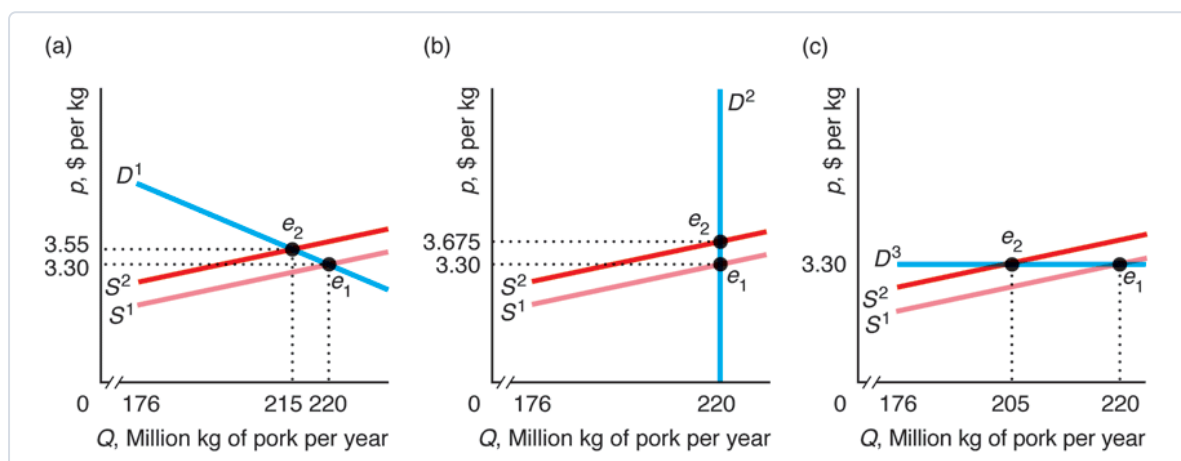
- **Elasticity** — the % change in one variable ÷ % change in another (a unit-free number).
- **Price elasticity of demand ( $E_D$ )** — % change in quantity demanded ÷ % change in own price (negative).
- **Elastic / inelastic / unit-elastic** —  $|E| > 1$  /  $|E| < 1$  /  $|E| = 1$ .
- **Point-elasticity trick** —  $E_D = (dQ/dp) \cdot (p/Q)$  ; on a linear curve elasticity **changes along the curve** (constant-elastic curves don't).
- **Cross-price elasticity** — substitutes (+) vs complements (-).
- **Income elasticity** — normal (+) vs inferior (-); **Engel curves / Engel's law**.
- **Supply elasticity** — % change in quantity supplied ÷ % change in price (positive).
- **Tax wedge & tax incidence** —  $p_b = p_s + t$  ; who bears the burden depends on **relative elasticities**; legal  $\neq$  economic incidence; unit vs ad-valorem tax; **deadweight loss**.

## Section-by-section main points

## Why do we care about elasticity? (deck)

**Core:** the **shape** of demand/supply curves determines **how much a shift in one moves the equilibrium** — and "shape" is summarised by **elasticity**. A steep (inelastic) curve and a flat (elastic) curve respond to the *same* shock very differently.

- **The motivating questions (deck):** Do "sin taxes" (cigarettes, sugar) hurt the poor? Why do firms **raise prices** even when it looks crazy (Netflix, Amazon Prime)? To what extent are **taxes on producers "passed on"** to consumers? Every answer runs through elasticity.
- **The visual intuition:** how far a **supply shift** moves price vs quantity depends entirely on the **slope/elasticity of demand** — see the figure below (perfectly inelastic demand → only price moves; perfectly elastic demand → only quantity moves).



**How to read it (why elasticity matters).** All three panels apply the **same supply shift**  $S^1 \rightarrow S^2$ . The **only** thing that differs is the **demand curve's shape**:

- **(a) Ordinary downward-sloping demand:** the shock splits into **both** a higher price *and* a lower quantity.
- **(b) Perfectly inelastic demand (vertical):** quantity can't respond, so the shock lands **entirely on price** (3.30 → 3.675). *This is the "necessity with no substitutes" case.*
- **(c) Perfectly elastic demand (horizontal):** price can't move, so the shock lands **entirely on quantity** (220 → 205).
- **Take-away:** the **more inelastic** the demand, the more a shock (or tax) shows up as **price**; the **more elastic**, the more it shows up as **quantity**. *This one picture is the whole reason we measure elasticity.*

## ★ Price elasticity of demand — definition

**Core:** the **price elasticity of demand** measures how sensitive quantity demanded is to the good's own price — the **% change in quantity ÷ % change in price**. It is a **pure (unit-free) number** and is **negative** (demand slopes down: price up → quantity down).

### PRICE ELASTICITY OF DEMAND

$$E_D = ( \Delta Q / Q \times 100 ) \div ( \Delta P / P \times 100 ) = \% \Delta Q \div \% \Delta P$$

*Reads:* how responsive quantity demanded is to a price change — "a 1% rise in price  $p$  leads to an  $E_D$  % change in quantity  $q$ ." —  $\Delta Q/Q \times 100$  = the percentage change in quantity demanded ·  $\Delta p/p \times 100$  = the percentage change in price ·  $E_D$  = the ratio (a unit-free number, negative for a normal good). The  $\times 100$ s cancel, so it's just % $\Delta Q$  over % $\Delta p$ .

- **Why unit-free matters (the point of the whole concept):** raw slopes aren't comparable — "1,000 tons less rice" means something different in **Vietnam** vs **Bhutan**, and depends on the currency. A percentage measure is **comparable across markets, currencies and time**.
- **Why negative:** demand curves slope **down**, so  $\Delta Q$  and  $\Delta p$  have opposite signs → the ratio is negative. (When people say a "higher" elasticity they usually mean a bigger **absolute value**  $|E_D|$ .)

## Computing elasticity — and the point-elasticity trick

**Core:** plug the two percentage changes into the formula. **On a linear demand curve the elasticity is not constant** — it depends on *where* you evaluate it, via  $E_D = (dQ/dp) \cdot (p/Q)$ .

- **Simple example (textbook Table 6.2):** quantity falls  $6 \rightarrow 4$  while price rises  $1 \rightarrow 2$ . Then

WORKED — SIMPLE EXAMPLE

$$E_D = ((4-6)/6) \div ((2-1)/1) = (-2/6) \div (1) = -1/3$$

*Reads:* a demand that is **inelastic** —  $(4-6)/6 = -0.33$  = the % fall in quantity (−33%) ·  $(2-1)/1 = 1$  = the % rise in price (+100%) ·  $-1/3$  = quantity falls **only 1/3 as fast** as price rises → **inelastic** demand.

- **A policy-style example (soda, Allcott et al. 2019):** own-price elasticity of sugary soda  $\approx -1.37$ . If price rises 5%, quantity changes by  $E_D \times 5\% = -1.37 \times 5 = -6.85\%$  (a ~7% fall). To cut demand 15%, price must rise  $-15 \div -1.37 \approx 11\%$ . → *this is exactly the sum a policymaker designing a soda tax needs.*
- **The point-elasticity trick (linear demand):** rewrite the formula as

POINT ELASTICITY

$$E_D = (\Delta Q/\Delta p) \cdot (p/Q) = (dQ/dp) \cdot (p/Q)$$

*Reads:* elasticity = **slope of the demand function** × the **price-to-quantity ratio at your chosen point** —  $dQ/dp$  = the **slope** (how many units  $Q$  changes per €1 of price; constant for a straight line) ·  $p/Q$  = the **price-to-quantity ratio** where you evaluate it (this is what varies along the curve).

- **Worked ( $Q = 10 - 2p$ ):** here  $dQ/dp = -2$  (always). But the **elasticity varies**:
  - at  $p = 2.5$  (so  $Q = 5$ ):  $E_D = -2 \times 2.5/5 = -1$  (**unit elastic**).
  - at  $p = 4$  (so  $Q = 2$ ):  $E_D = -2 \times (p/Q) = -2 \times 4/2 = -4$  → **elastic**. ( $\triangle$  The textbook misprints this as  $-2 \times 2/2 = -2$  — it accidentally puts  $Q=2$  in the numerator instead of  $p=4$ ; the correct  $p/Q$  is  $4/2 = 2$ , so  $E_D = -4$ .)
  - **The pattern: demand is more elastic at higher prices** — at  $p=2.5$  it's unit-elastic (−1), at  $p=4$  it's elastic (−4). A fixed unit change is a bigger *percentage* change when  $Q$  is small and  $p$  is large.

- **Constant-elasticity demand (contrast):** the form  $Q = a \cdot p^\epsilon$  has the **same** elasticity  $\epsilon$  at *every* price (used in empirical estimation via  $\ln Q = \ln a + \epsilon \cdot \ln p + \text{error}$ ). So: **linear** → **elasticity changes along the curve**; **constant-elastic** → **it doesn't**.

## Elastic, inelastic, unit-elastic — and why it matters

**Core:** we **name** the ranges of elasticity, and each has a sharp economic meaning for **how supply shocks split into price vs quantity**. | Name | Level (of  $E_D$ ) | Meaning | Curve looks... | |---|---|---|---| | **Inelastic** |  $-1 < E_D < 0$  ( $|E| < 1$ ) |  $\% \Delta Q$  **smaller** than  $\% \Delta p$  | relatively **steep** | | **Unit elastic** |  $E_D = -1$  |  $\% \Delta Q$  **equals**  $\% \Delta p$  | — | | **Elastic** |  $E_D < -1$  ( $|E| > 1$ ) |  $\% \Delta Q$  **bigger** than  $\% \Delta p$  | relatively **flat** |

- **Why it matters (the punchline):** with **inelastic** demand, **supply shocks translate into big price changes** (little quantity response) — critical for e.g. **cocoa** growers in Ghana/Côte d'Ivoire (world cocoa elasticity  $\approx -0.19$  to  $-0.96$ ; policy point estimate  $-0.34$ ). With **elastic** demand, shocks translate into **quantity** changes.
- **What makes demand elastic — substitutes.** The single biggest driver is **availability of close substitutes**: pharmaceuticals with few substitutes are **inelastic** (people pay whatever for a needed medicine); goods with many substitutes are **elastic**.
- **Aggregation level matters:** a *specific brand* is more **elastic** than the *category* (more substitutes for one brand). US beer: **product-level** median elasticity  $\approx -4.74$ , but beer **overall**  $\approx -0.60$ .

## Cross-price elasticity — substitutes vs complements

**Core:** the **cross-price elasticity** measures how demand for good **A** responds to the price of a *different* good **B** — and its **sign** tells you the relationship.

### CROSS-PRICE ELASTICITY

$$E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B \rightarrow > 0 = \text{substitutes} \cdot < 0 = \text{complements}$$

**Reads:** how the quantity of **A** responds to a price change in **B** —  $\% \Delta Q_A$  = percentage change in demand for good A ·  $\% \Delta p_B$  = percentage change in the price of good B · **sign: positive** → **substitutes** (Heineken dearer → more Carlsberg bought), **negative** → **complements** (phones dearer → fewer phone cases). *The larger the positive value, the closer the substitutes.*

- **Why it's used:** gauging **competitive pressure** and evaluating **mergers** (competition authorities ask how closely two products substitute). (*Deck aside: cross-price elasticity of cigarettes & alcohol  $\approx -1$  — strong complements, Krauss et al. 2014.*)

## Income elasticity — normal vs inferior goods, Engel curves

**Core:** the **income elasticity** measures how demand responds to **income** — its **sign** distinguishes **normal** from **inferior** goods.

#### INCOME ELASTICITY

$$E_{income} = \% \Delta Q \div \% \Delta I \rightarrow > 0 = \text{normal good} \cdot < 0 = \text{inferior good}$$

*Reads:* how quantity demanded responds to an income change —  $\% \Delta Q$  = percentage change in quantity ·  $\% \Delta I$  = percentage change in income · **sign: positive → normal** (demand rises with income), **negative → inferior** (demand falls as income rises — e.g. instant noodles, store-brand goods).

- **Nuance:** a good can be **normal at low incomes and inferior at higher ones** (hostels for a student: more nights as income rises, then switch to hotels). Tastes vary by person.
- **Uses:** long-term **projections** (healthcare, appliances in low-income countries) and predicting the **business cycle** (pharma is income-insensitive → "defensive stock").
- **Engel curves / Engel's law (Ernst Engel, 1800s Prussian statistician):** plot demand (vertical) against income (horizontal) — **positive slope for normal goods, negative for inferior**. **Engel's law:** the share of income spent on food *falls* as income rises → the income elasticity of food is **less than 1**.

#### Supply elasticity — same principle

**Core:** the **price elasticity of supply** measures how sensitive quantity *supplied* is to price — same formula, but **positive** (supply slopes up).

#### PRICE ELASTICITY OF SUPPLY

$$E_S = \% \Delta Q_S \div \% \Delta p \quad (\text{positive})$$

*Reads:* how responsive quantity supplied is to a price change —  $\% \Delta Q_S$  = percentage change in quantity supplied ·  $\% \Delta p$  = percentage change in price · **E<sub>S</sub>** = the ratio (**positive**: higher price → more supplied). A **steep** supply curve is **inelastic** (hard to expand output quickly); a **flat** one is **elastic**.

#### Elasticity over time — short run vs long run

**Core:** demand (and supply) are usually **more elastic in the long run** than the short run, because people have **more time to adjust / find substitutes**.

- **Electricity (Labandeira et al. 2017):**  $\approx -0.24$  short-run vs  $-0.6$  long-run — over time consumers buy more efficient equipment, change habits.
- **The deck's cases:** a drop in **gas prices** does little in the short run but more in the long run; likewise **computers**. *Exam framing:* "in the short run demand is inelastic (people are locked in); given time they substitute, so long-run demand is more elastic."

#### ★ Taxes on goods & services — the wedge, incidence & equivalence (deck; textbook §7.5)

**Core:** a tax **drives a wedge** between the price **buyers pay** ( $p_b$ ) and the price **sellers receive** ( $p_s$ ):  $p_b = p_s + t$ . It raises the buyer price, lowers the seller price, cuts the quantity traded, and creates a

**deadweight loss.** The split of the burden — **tax incidence** — depends on the **relative elasticities**, and (under perfect competition) **not** on who is legally taxed.

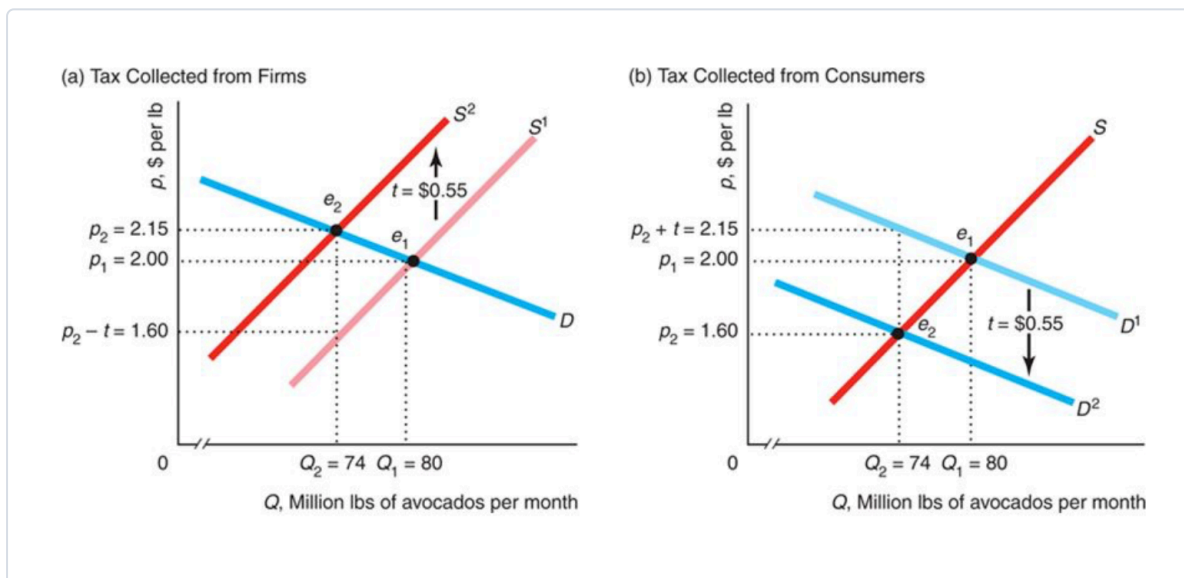
- **Two types of tax (deck):**

- **Unit (specific) tax** — a fixed **amount  $t$  per unit** (per litre of petrol, per cigarette).
- **Ad valorem tax** ("sales tax"/VAT) — a **fraction  $\alpha$  of the price/spend** (government keeps a share of each euro spent).

- **The wedge (worked unit-tax example,  $t = €3$ ):** with  $Q_D = 23 - p_b$ ,  $Q_S = -1 + p_s/2$ , and  $p_b = p_s + 3$ , solving gives  $p_s = 14$ ,  $p_b = 17$ ,  $Q = 6$ . Buyers pay **€3 more** than sellers receive; quantity falls from the no-tax equilibrium.

- **Tax revenue** =  $t \times Q = 3 \times 6 = €18$ . The rest of the surplus lost is **deadweight loss** (here  $3 \times (7-6)/2 = €1.5$ ) — trades that *would* have happened but now don't.

- **Legal incidence  $\neq$  economic incidence (the key result).** It **doesn't matter** whether the tax is collected from **firms** (supply shifts up by  $t$ ) or **consumers** (demand shifts down by  $t$ ) — the quantity, the prices actually paid/received, and the DWL are **identical**. *All that matters is that the tax drives a wedge.* (Perfect-competition assumptions make this exact: no transaction costs, perfect information.)



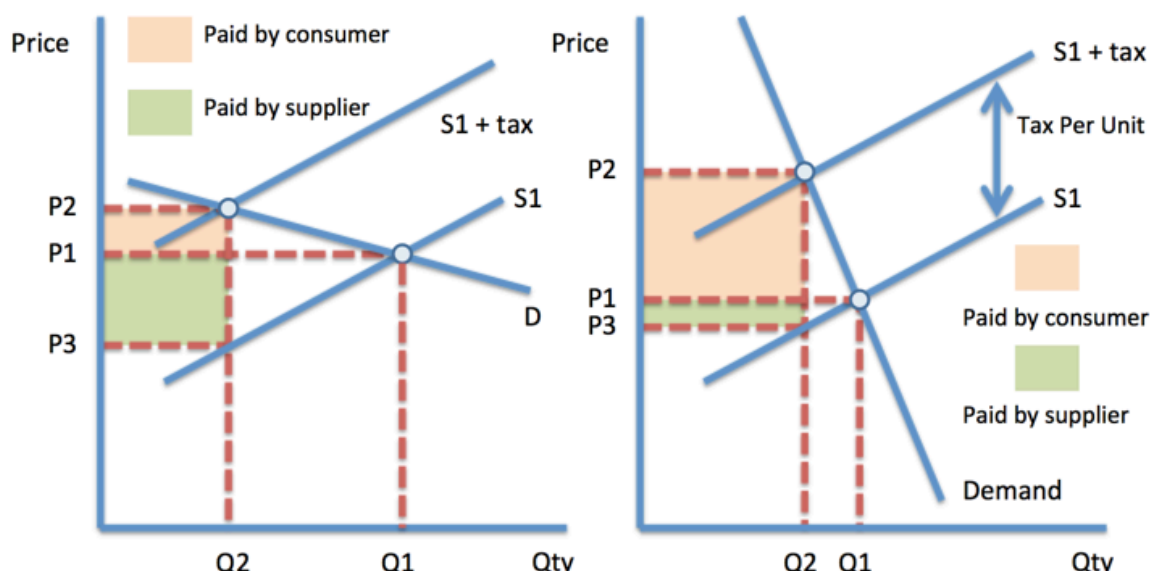
**How to read it (tax equivalence).** Both panels impose the **same \$0.55 tax** on avocados. **(a)** collects it from **firms** → the supply curve shifts **up** by  $t$ ; **(b)** collects it from **consumers** → the demand curve shifts **down** by  $t$ . In **both** cases the result is identical:  **$Q$  falls to 74, buyers pay 2.15, sellers receive 1.60**, and the wedge between them is exactly **\$0.55**. → *The economics doesn't care who writes the cheque.*

- **What *does* decide the split: relative elasticities.** The **more inelastic** side of the market bears **more** of the tax (it can't escape by changing quantity). The **more price-sensitive (elastic)** side is affected **less** on price, **more** on quantity.

## Indirect Taxes with Different Coefficient of PED

If the co-efficient of price elasticity of demand  $>1$ , then most of the burden of an indirect tax will be absorbed by the supplier

If the co-efficient of price elasticity of demand  $<1$ , most of an indirect tax can be passed on to the final consumer



**How to read it (incidence & elasticity).** The tax raises the supply curve to  $S + \text{tax}$ . The **new higher price**  $P2$  is what **consumers** pay;  $P3$  is what **suppliers** keep. The split of the rectangle between them is the **incidence**:

- **Left — elastic demand ( $PED > 1$ ):** consumers easily cut back, so firms **can't raise the price much** → the **supplier absorbs most** of the tax.
- **Right — inelastic demand ( $PED < 1$ ):** consumers keep buying regardless, so the firm **passes most** of the tax through → the **consumer bears most** of it.
- **Policy take-aways:** (1) "sin taxes" on inelastic goods (cigarettes, alcohol) fall mostly on **consumers** and raise a lot of revenue with **little DWL** (quantity barely moves) — which is *why* governments like them, and why they can be **regressive** (hit the poor). (2) We **prefer taxes on inelastic bases** — they distort choices least.
- **Subsidies are the mirror image:**  $p_s = p_b + s$  — a subsidy lowers the buyer price, raises the seller price, **expands** quantity above the efficient level, and also creates a **deadweight loss** (units produced that cost more than consumers value them). Its incidence, too, depends on elasticities (inelastic demand — e.g. emergency care, primary education — → small DWL).

## Cases & examples

Read each as a worked use of an elasticity number (or a tax result).

- **Sugary-soda tax (Allcott et al. 2019).** Own-price elasticity  $\approx -1.37$  (elastic). A 5% price rise →  $\sim 7\%$  fall in quantity; to cut demand 15% you'd raise price  $\sim 11\%$ . → *Illustrates: elasticity as the design tool for a corrective ("sin") tax.*

- **World cocoa demand.** Inelastic ( $\approx -0.19$  to  $-0.96$ ; policy estimate  $-0.34$ ). → *Illustrates: **inelastic demand** → **supply shocks hit price**, so bad harvests swing incomes for growers in Ghana / Côte d'Ivoire.*
- **US beer — brand vs category (Miller et al. 2021).** Product-level elasticity  $\approx -4.74$ , category  $\approx -0.60$ . → *Illustrates: **aggregation** — a single brand has close substitutes (elastic), the category doesn't (inelastic).*
- **Sweden–Denmark cross-border alcohol (Asplund et al. 2007).** Swedes drive to Denmark for cheaper drink. Estimated for spirits: **own-price**  $-1.3$  (elastic), **cross-price (Danish price)**  $+0.3$  (substitute), **income**  $+1.4$  (normal good). → *Illustrates: **all three elasticities at once** — own, cross, income — from real data.*
- **Electricity demand (Labandeira et al. 2017).**  $\approx -0.24$  short-run vs  $-0.6$  long-run. → *Illustrates: demand is **more elastic in the long run** (time to adjust equipment/habits).*

## Formula sheet — quick reference

| Quantity                          | Formula                                                             | Sign / meaning                                   |
|-----------------------------------|---------------------------------------------------------------------|--------------------------------------------------|
| <b>Price elasticity of demand</b> | $E_D = \% \Delta Q_D \div \% \Delta p$                              | <b>negative;</b>                                 |
| <b>Point elasticity (linear)</b>  | $E_D = (dQ/dp) \cdot (p/Q)$                                         | varies along the curve; more elastic at higher p |
| <b>Constant-elastic demand</b>    | $Q = a \cdot p^\epsilon \rightarrow \ln Q = \ln a + \epsilon \ln p$ | elasticity = $\epsilon$ everywhere               |
| <b>Cross-price elasticity</b>     | $E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B$               | <b>+ substitutes, – complements</b>              |
| <b>Income elasticity</b>          | $E_{\text{income}} = \% \Delta Q \div \% \Delta I$                  | <b>+ normal, – inferior;</b> food $< 1$ (Engel)  |
| <b>Supply elasticity</b>          | $E_S = \% \Delta Q_S \div \% \Delta p$                              | <b>positive</b>                                  |
| <b>Tax wedge</b>                  | $p_b = p_s + t$                                                     | buyers pay more, sellers get less; DWL           |
| <b>Tax revenue</b>                | $t \times Q_{\text{taxed}}$                                         | rectangle; rest of loss = deadweight loss        |
| <b>Subsidy wedge</b>              | $p_s = p_b + s$                                                     | expands Q above efficient; also a DWL            |

## Exam pointers

- Compute an elasticity from two (p, Q) points and **name** it (elastic/inelastic/unit) — the **−1/3 example**.
- Use the **point-elasticity trick**  $E_D = (dQ/dp) (p/Q)$  on a **linear demand** and show elasticity **changes along the curve** (more elastic at higher price).
- Give the **sign rules** for **cross-price** (substitutes/complements) and **income** (normal/inferior) elasticities, with an example each.
- Solve a **unit-tax market** ( $p_b = p_s + t$ ) for  $p_b$ ,  $p_s$ ,  $Q$ , compute **revenue** and **deadweight loss** — the **t=€3 example**.
- State and **explain** the two big tax results: **legal  $\neq$  economic incidence** (doesn't matter who's taxed) and **incidence depends on relative elasticities** (inelastic side pays more); use the **sin-tax**

application.

- Explain **why elasticity is unit-free** and why demand is **more elastic in the long run** and for **narrower product categories**.

**Bridge.** Lecture 3 turns the Chapter 2 model from *qualitative* ("the curve shifts left") into *quantitative* ("by how much, and split how between price and quantity"). Elasticity is also the hinge to later chapters: **firm pricing** (a monopolist's markup depends on  $E_D$ ), **tax policy & deadweight loss** (Ch. 7), and **who gains from trade**.

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## ★ Fun facts & memorable details (Lecture 3)

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Sticky bits from Elasticities & Taxes.

- **It doesn't matter who you tax.** In a competitive market the burden of a tax is **identical** whether the government bills the shop or the shopper — "all that matters is that the tax drives a wedge." (This is why arguing about *who* a tax is "on" often misses the point.)
- **Why cigarette & alcohol taxes raise so much money:** demand is **inelastic**, so people keep buying — the tax is **passed to consumers** with little drop in quantity (and is often **regressive**).
- **The soda-tax number:** the own-price elasticity of sugary drinks ( $\approx -1.37$ ) is a real input to real policy — a 5% price rise cuts consumption  $\sim 7\%$ .
- **Engel's law (1857):** the **richer** you get, the **smaller** the share of income you spend on **food** — one of the oldest empirical regularities in economics, and still used to gauge living standards.
- **Pharma = "defensive stock":** demand for medicine barely moves with income (income-inelastic), so drug companies hold up in recessions.
- **Elasticity has a time dimension:** when petrol jumps, you can't do much *this week* (inelastic), but over years you buy a smaller car and move closer to work (elastic) — the "short-run vs long-run" split.
- **Brand vs category:** demand for *a* beer is wildly elastic ( $\approx -4.7$  — switch brands!), but demand for *beer* is inelastic ( $\approx -0.6$ ) — same drink, opposite elasticity, just different zoom level.
- **Swedes really do drive to Denmark for cheap booze** — enough that economists measured the cross-border **cross-price elasticity** of spirits ( $+0.3$ ) from the data (Asplund et al. 2007).